Glycolysis

Metabolism

- **Metabolism is "the entire set of enzyme-catalyzed transformations of organic molecules in living cells.**
- **Two broad classes:**
- **Catabolism & Anabolism**
- **Catabolic Pathways:**
- **Transform fuels into cellular energy**

Anabolic Pathways

- **Requires inputs of energy to proceed.**
- **Useful energy + small molecules complex molecules.**
- **Pathways that can be either anabolic or catabolic, depending on the energy conditions in the cell are referred to as amphibolic pathways**

Glycolysis

- Glycolysis occurs in almost every living cell.
- It was the first metabolic sequence to be studied.
- This pathway is also called Embden-Meyerhof pathway (E.M-Pathway).
- It occurs in cytosol.

Definition

- Glycolysis is defined as the **sequence of reactions converting glucose to pyruvate or lactate, with the production of ATP**.
- **Salient features:**
- Takes place in **all cells** of the body.
- The enzymes of this pathway are present in the **cytosomal fraction** of the cell.

- Glycolysis occurs in the absence of oxygen (anaerobic) or in presence of oxygen (aerobic).
- Lactate is the end product under anaerobic condition.
- In aerobic condition, pyruvate is formed, which is then oxidized to CO_2 & H_2O.

- Glycolysis is a major pathway for ATP synthesis in tissues lacking mitochondria, erythrocytes, cornea, lens etc.
- Glycolysis is very essential for brain which is dependent on glucose for energy.

- The glucose in brain has to undergo glycolysis before it is oxidized to CO_2 & H_2O.
- Glycolysis is a central metabolic pathway with many of its intermediates providing branch point to other pathways.
- The intermediates of glycolysis are useful for the synthesis of amino acids and fat.

Glucose entry into cells

- **Glucose transporter-4 (GluT4) transports glucose from extracellular fluid to muscle cells & adipocytes.**
- **This is under the influence of Insulin.**
- **In diabetes mellitus, insulin deficiency hinders the entry of glucose into the peripheral cells.**
- **GluT2 is the transporter in liver cell.**
- **It is not under the control of insulin.**

Reactions of Glycolysis

- Divided into three distinct phases.
- Energy investment phase or priming phase
- Splitting phase
- Energy generation phase.

Energy investment phase

- Glucose obtained from
- The diet through intestinal hydrolysis of lactose, sucrose, glycogen, or starch is brought into the hexose phosphate pool through the action of hexokinase.
- Free glucose is phosphorylated to glucose 6 phosphate by hexokinase

- Hexokinase splits the ATP into ADP & Pi, the Pi is added to the glucose.
- Hexokinase Hexokinase is a key glycolytic enzyme.



- **Phosphorylated sugar molecules do not readily penetrate cell membranes without specific carriers, this commits glucose to further metabolism in the cell.**
- **In all tissues, the phosphorylation of glucose is catalyzed by hexokinase, one of the three regulatory enzymes of glycolysis.**

Hoxokinase and glucokinase

	Hexokinase	Glucokinase
Occurrence	In all tissues	Only in liver
Km Value	10^{-2} mmol/L	20 mmol/L
Affinity to substrate	High	Low
Specificity	Acts on glucose, fructose and mannose	Acts only on glucose
Induction	Not induced	Induced by insulin & glucose
Function	Even when blood sugar level is low, glucose is utilized by body cells	Acts only when blood glucose level is more then 100 mg/dl; then glucose is taken up by the liver cells for glycogen synthesis

Isomerization of Glucose 6-P

- Glucose 6 P is a central molecule with a variety of metabolic fates- glycolysis, glycogenesis, gluconeogenesis and HMP pathway.
- The isomerization of Glucose 6-P (an aldose sugar) to Fructose 6-P (a ketose sugar) is catalyzed by phosphohexose isomerase
- It requires Mg^{+2} ions.

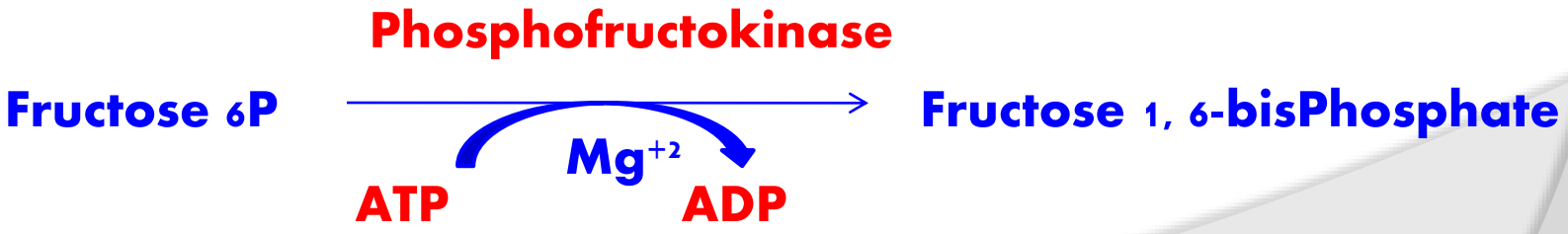
- **The reaction is readily reversible, is NOT a rate limiting or regulated step.**

Phosphohexose isomerase & Mg^{+2}

Glucose 6-Phosphate $\longleftrightarrow$ **Fructose 6-Phosphate**

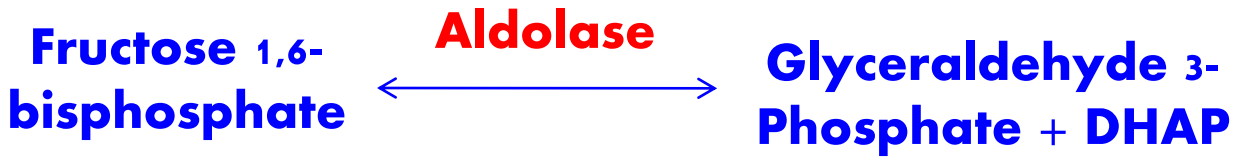
Phosphorylation of Fructose 6-P

- Fructose 6- phosphate is phosphorylated to Fructose 1, 6- bisphosphate by Phosphofructokinase (PFK)
- The PFK reaction is the rate-limiting step.
- It is controlled by the concentrations of the substrates ATP & Fructose 6-P



Splitting Phase

- The six carbon Fructose 1, 6- bisphosphate is split to 2 three carbon compounds.
- Glyceraldehyde 3- phosphate & Dihydroxy acetone phosphate by the enzyme aldolase (Fructose 1, 6- bisphosphate aldolase).
- The reaction is reversible is not subject to regulation.



Isomerization of DHAP

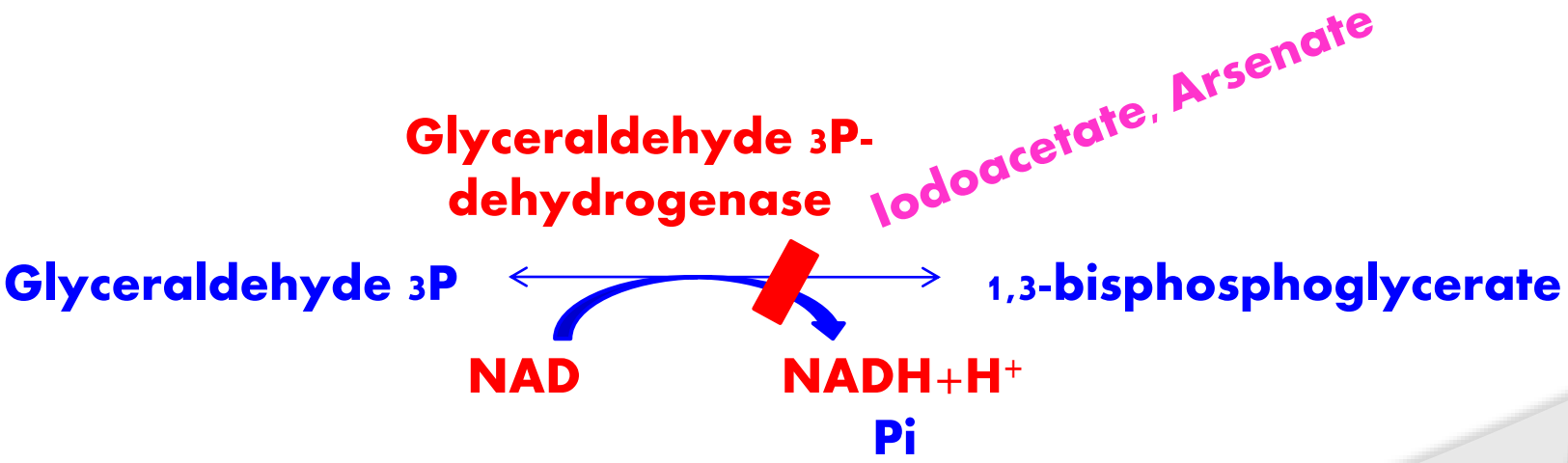
- **Phosphotriose isomerase catalyzes the reversible interconversion of dihydroxyacetone phosphate & glyceraldehyde 3-phosphate.**
- **Two molecules of glyceraldehyde 3-phosphate are obtained from one molecule of glucose.**

DHAP ⟷ (**Phosphohexose isomerase**) **Glyceraldehyde 3-Phosphate**

Oxidation of glyceraldehyde 3P

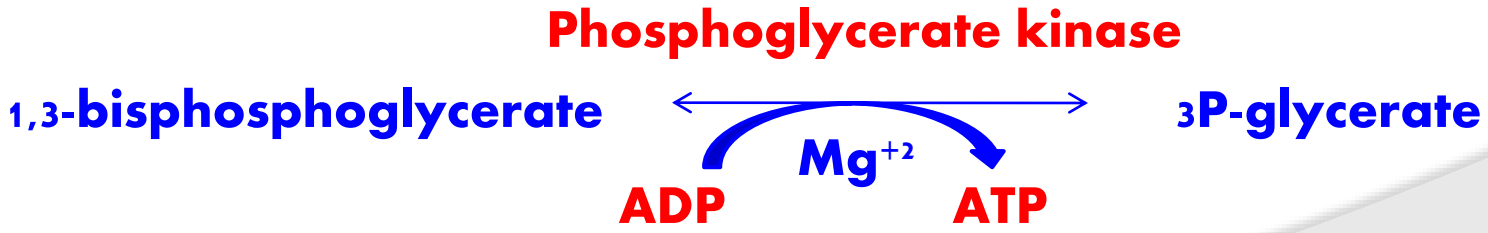
- Glyceraldehyde 3-phosphate dehydrogenase converts Glyceraldehyde 3-phosphate to 1,3-bisphosphoglycerate.
- This step is important as it is involved in the formation of NADH $+H^+$ & a high energy compound 1,3- bisphosphoglycerate.

- **In aerobic condition, NADH passes through the ET C and 6 ATP are synthesized by oxidative phosphorylation.**



Formation of ATP from 1,3-bisphosphoglycerate & ADP

- The enzyme phosphoglycerate kinase acts on 1,3- bisphosphoglycerate resulting in the synthesis of ATP and formation of 3-phosphoglycerate.



- This step is a substrate-level phosphorylation
- Production of a high-energy P is coupled to the conversion of substrate to product, instead of resulting from oxidative phosphorylation.
- The energy will be used to make ATP in the next reaction of glycolysis.

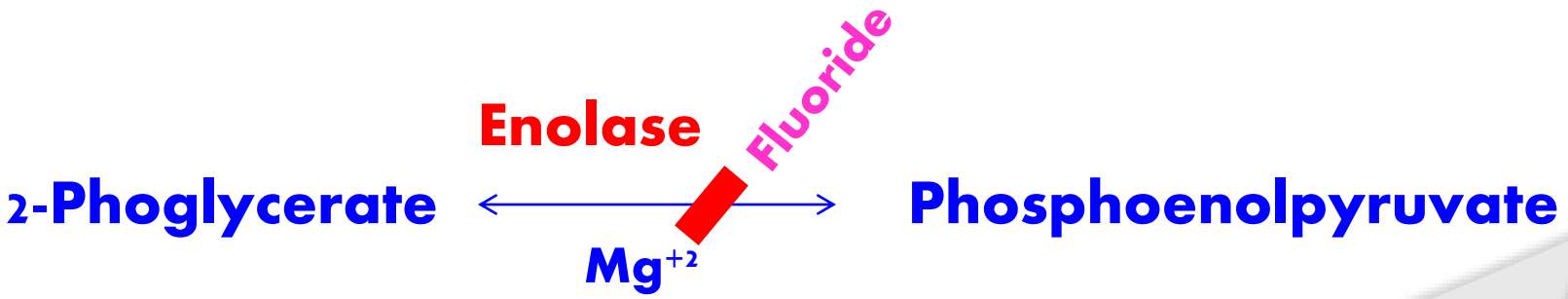
- **The formation of ATP by P group transfer from a substrate such as 1,3-bisphosphoglycerate is referred to as a substrate-level phosphorylation.**
- **Unlike most other kinases, this reaction is reversible.**

- **3- Phosphoglycerate is converted to 2-Phosphoglycerate by phosphoglycerate mutase**
- **This is isomerization reaction.**

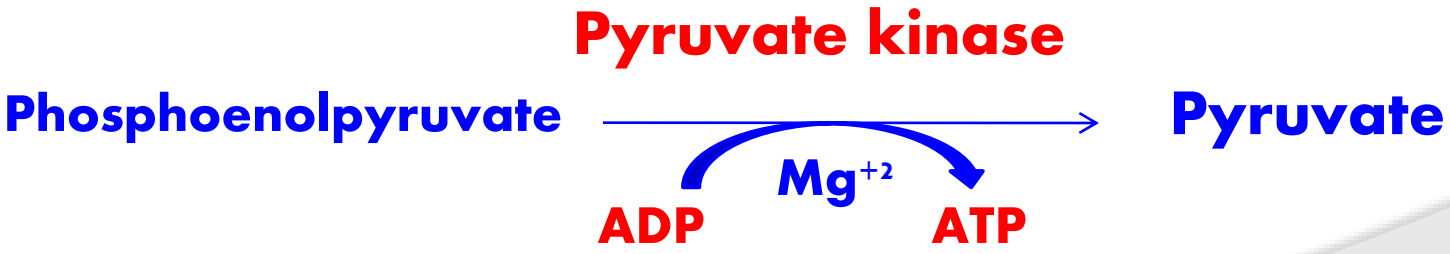
Phosphoglycerate mutase

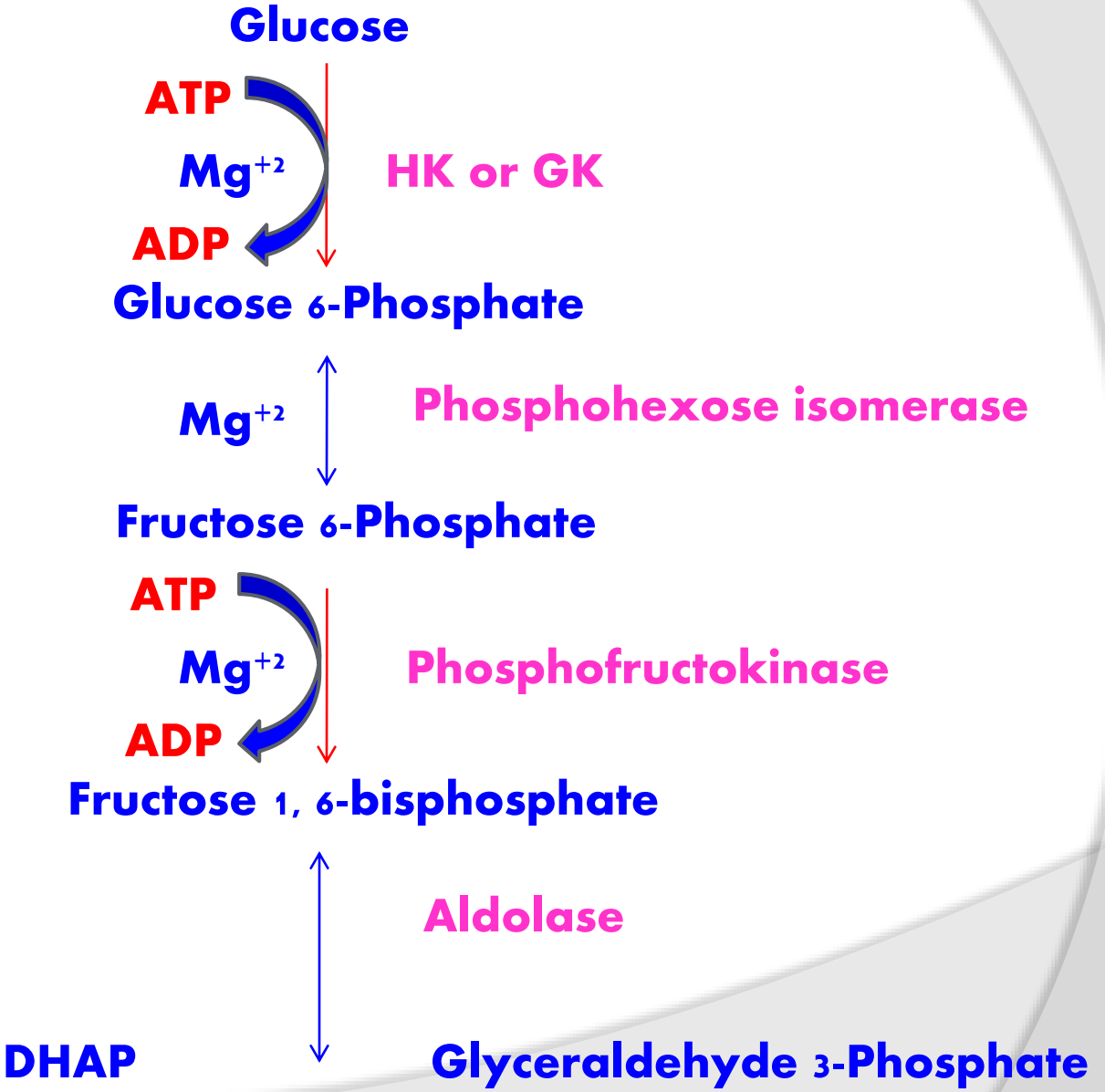
3-Phosphoglycerate ⟷ 2P-glycerate

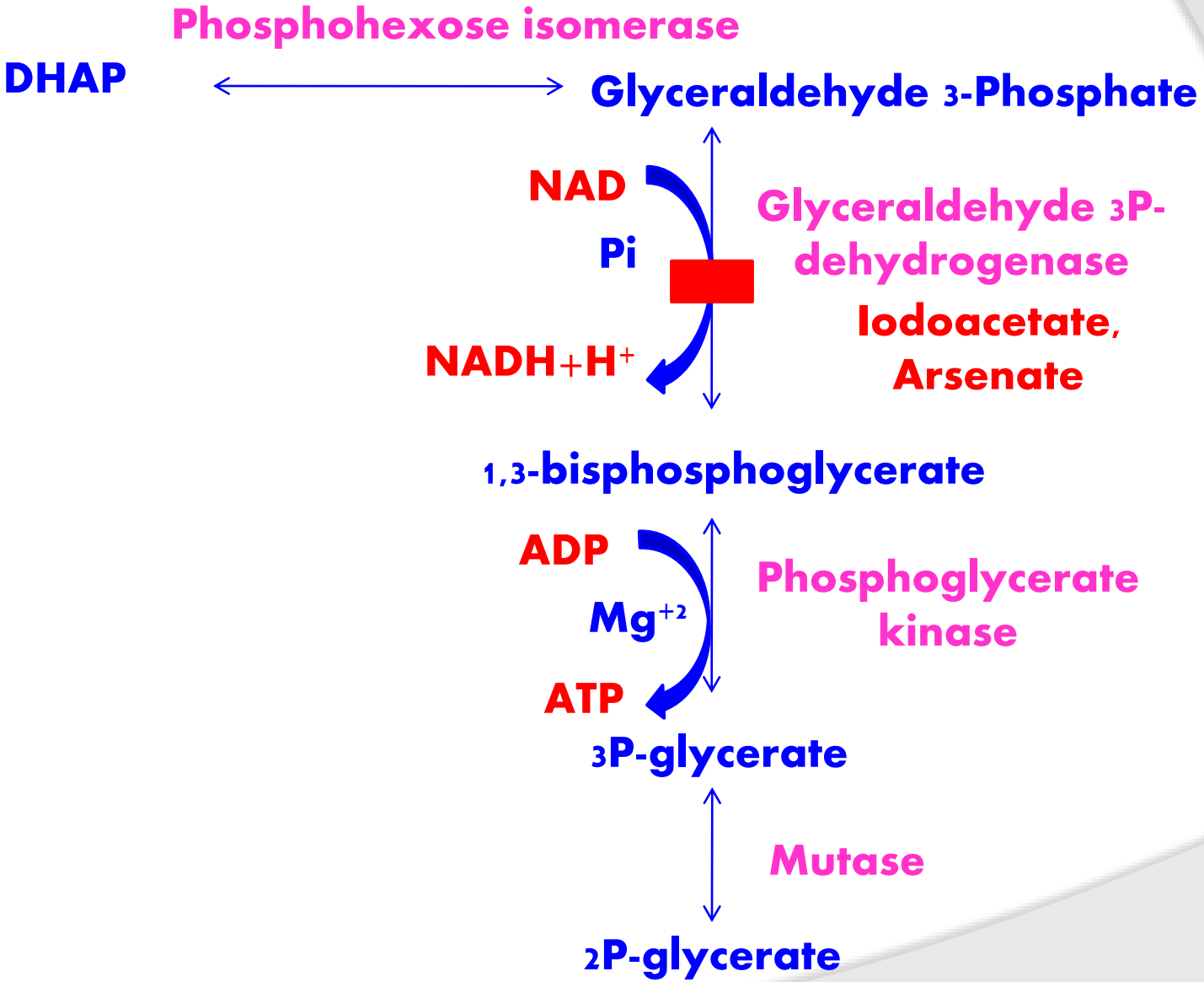
- The high energy compound PEP is generated from 2- Phosphoglycerate by the enzyme enolase.
- This enzyme requires Mg^{+2} or Mn^{+2} and is inhibited by fluoride.

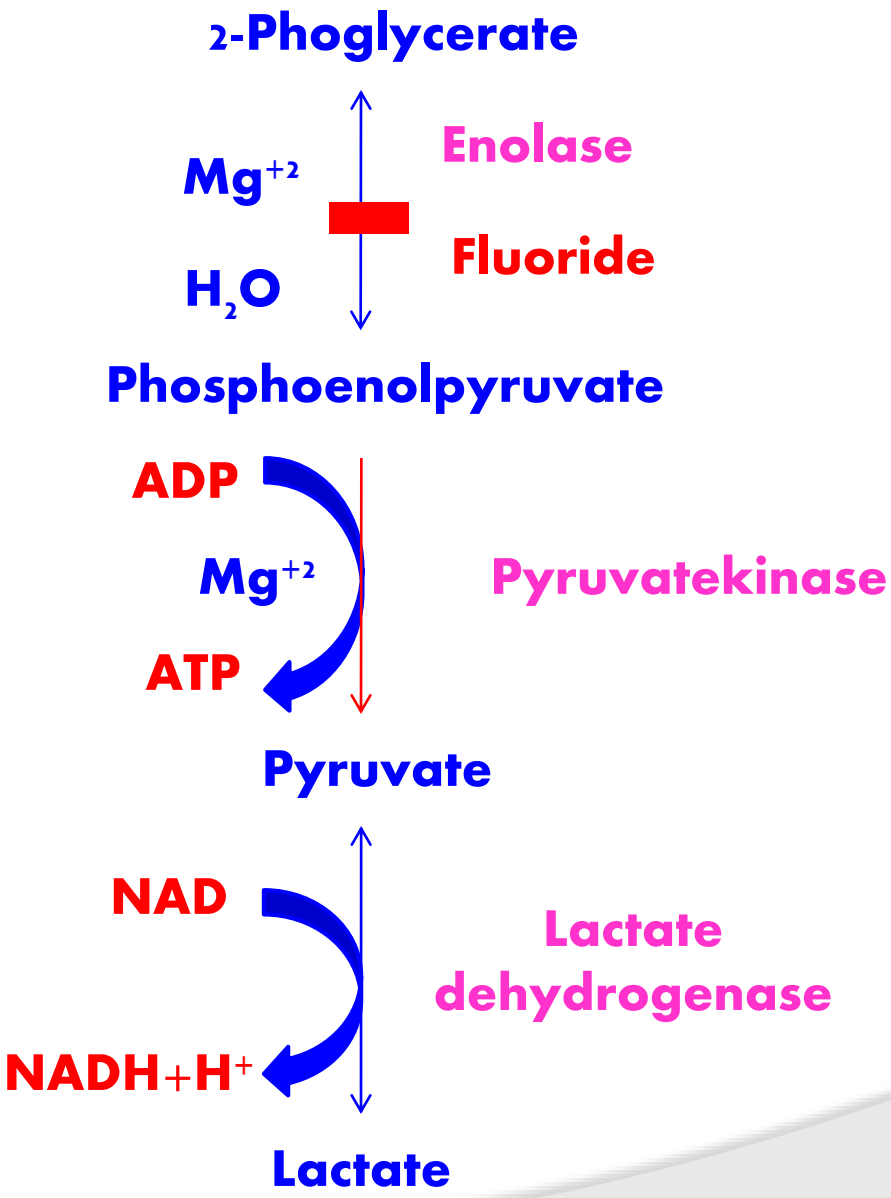


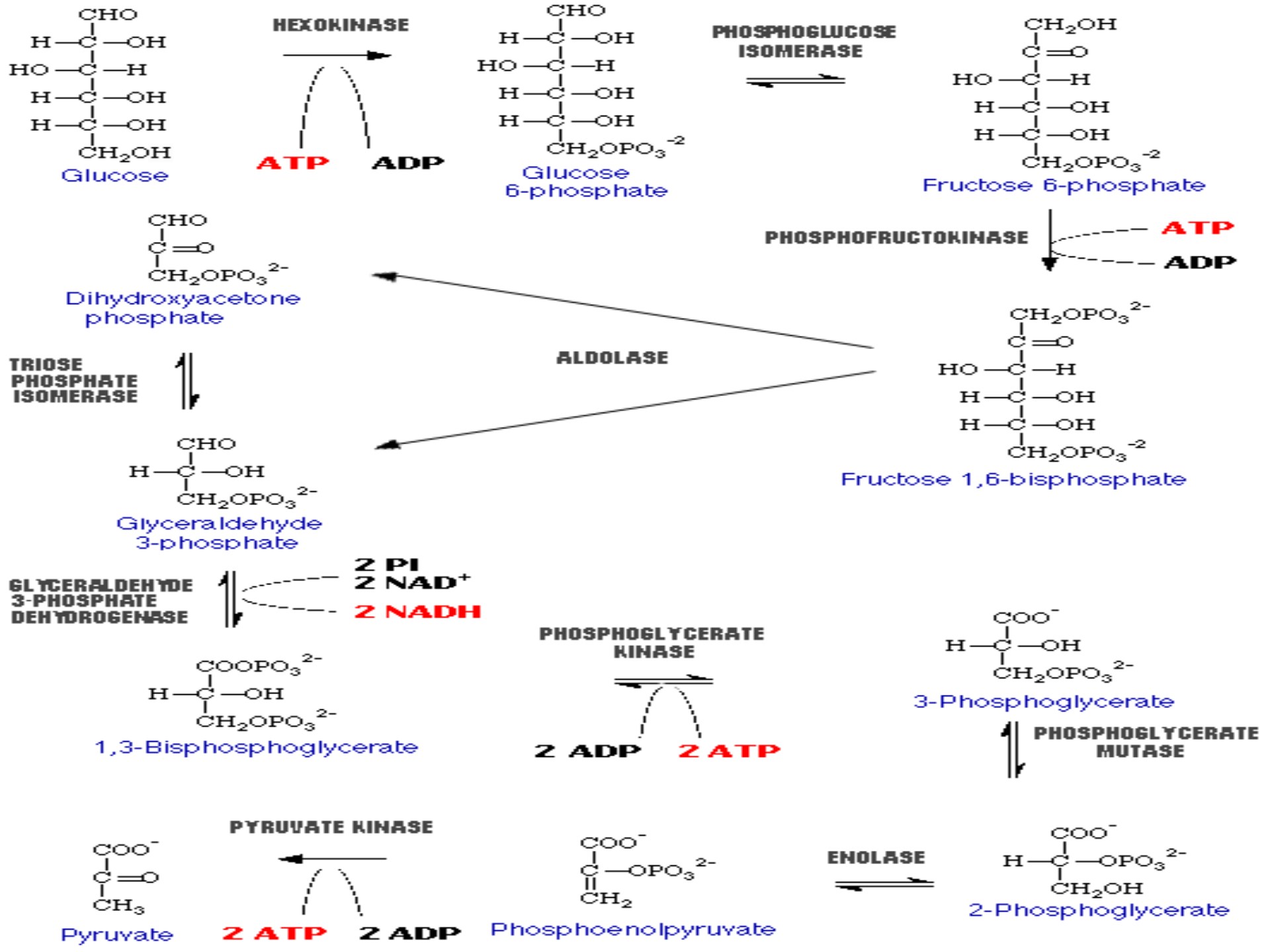
- The enzyme pyruvate kinase catalyses the transfer of high energy phosphate from PEP to ADP, leading to the formation of ATP.
- This step is also a substrate level phosphorylation.




Glucose
ATP
Mg+2
ADP
HK or GK
Glucose 6-Phosphate
Mg+2
Phosphohexose isomerase
Fructose 6-Phosphate
ATP
Mg+2
ADP
Phosphofructokinase
Fructose 1, 6-bisphosphate
Aldolase
DHAP
Glyceraldehyde 3-Phosphate


Phosphohexose isomerase
DHAP
Glyceraldehyde 3-Phosphate
NAD
Pi
NADH+H+
Glyceraldehyde 3P-dehydrogenase
Iodoacetate, Arsenate
1,3-bisphosphoglycerate
ADP
Mg+2
ATP
Phosphoglycerate kinase
3P-glycerate
Mutase
2P-glycerate


2-Phoglycerate
Mg+2
H2O
Enolase
Fluoride
Phosphoenolpyruvate
ADP
Mg+2
ATP
Pyruvatekinase
Pyruvate
NAD
NADH+H+
Lactate dehydrogenase
Lactate


Glucose
HEXOKINASE
ATP
ADP
Glucose 6-phosphate
PHOSPHOGLUCOSE ISOMERASE
Fructose 6-phosphate
PHOSPHOFRUCTOKINASE
ATP
ADP
Dihydroxyacetone phosphate
ALDOLASE
Fructose 1,6-bisphosphate
TRIOSE PHOSPHATE ISOMERASE
Glyceraldehyde 3-phosphate
GLYCERALDEHYDE 3-PHOSPHATE DEHYDROGENASE
2 PI
2 NAD+
2 NADH
1,3-Bisphosphoglycerate
PHOSPHOGLYCERATE KINASE
2 ADP
2 ATP
3-Phosphoglycerate
PHOSPHOGLYCERATE MUTASE
PYRUVATE KINASE
Pyruvate
2 ATP
2 ADP
Phosphoenolpyruvate
ENOLASE
2-Phosphoglycerate

Regulation of glycolysis

- Three regulatory enzymes:
- Hexokinase & glucokinase
- Phosphofructokinase
- Pyruvate kinase
- Catalysing the irreversible reactions regulate glycolysis.

Hexokinase

- **Hexokinase is inhibited by glucose 6-phosphate.**
- **This enzyme prevents the accumulation of glucose 6-phosphate due to product inhibition.**

Glucokinase

- **Glucokinase, which specifically phosphorylates glucose, is an inducible enzyme.**
- **The substrate glucose, probably through the involvement of insulin, induces glucokinase**

Phosphofructokinase (PFK)

- Phosphofructo kinase (PFK) is the most important regulatory enzyme in glycolysis
- PFK is an allosteric enzyme regulated by allosteric effectors ATP, citrate & H^+ ions (low pH) are the most important allosteric inhibitors.
- Fructose 2 ,6-bisphosphate, ADP, AMP & Pi are the allosteric activators.

- **Role of fructose 2,6-bisphosphate in glycolysis**
- **Fructose-2,6-bisphosphate (F2,6-BP) is considered to be the most important regulatory factor (activator) for controlling PFK & ultimately glycolysis in the liver.**
- **F2,6-BP is synthesized from fructose 6-p by the enzyme phosphofructokinase called PFK-2 (PFK-1 is the glycolytic enzyme)**

- **$F_{2,6}$-BP is hydrolysed by fructose 2,6 - bisphosphatase.**
- **The function of synthesis & degradation of $F_{2,6}$-BP is brought out by a single enzyme (same polypeptide with two active sites) which is referred to as bifunctional enzyme.**
- **The activity of PFK-2 & fructose 2,6- bisphosphatase is controlled by covalent modification which, in turn, is regulated by c AMP.**

- **Cyclic AMP brings about dephosphorylation of the bifunctional enzyme, resulting in inactivation of active site responsible for the synthesis of $F_{2,6}$-BP but activation of the active site responsible for the hydrolysis of $F_{2,6}$-BP**

Pyruvate kinase

- PK Inhibited by ATP & activated by $F_{1,6}$-BP.
- Pyruvate kinase is active (a) in dephosphorylated state & inactive (b) in phosphorylated state.
- Inactivation of pyruvate kinase is brought about by cAMP-dependent protein kinase.
- The hormone glucagon inhibits hepatic glycolysis by this mechanism.

Energy yield from glycolysis

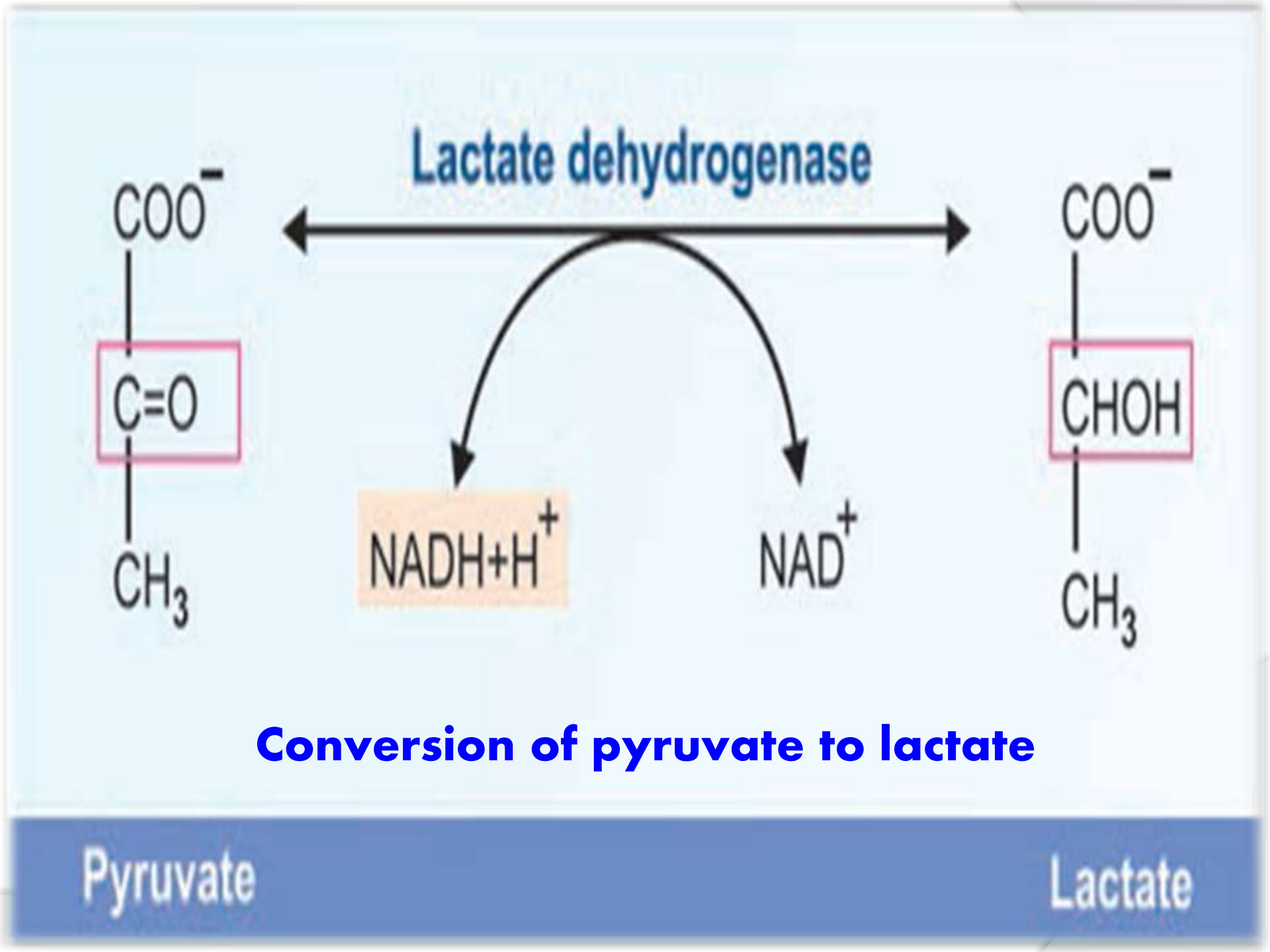
- During anaerobic:
- One molecule of glucose is converted to 2 molecules of lactate, there is a net yield of 2 molecules of ATP.
- 4 molecules of ATP are synthesized by 2 substrate level phosphorylation.
- 2 ATP molecules are used in steps 1 & 3,
- Hence, net yield is 2 ATP.

During Aerobic condition

- 2 NADH molecules, generated in the glyceraldehyde 3P-dehydrogenase reaction & enter ETC.
- NADH provides 3 ATP, this reaction generates 3x2=6 ATP
- Total ATP is 6+2=8 ATP.

Conversion of pyruvate to lactate

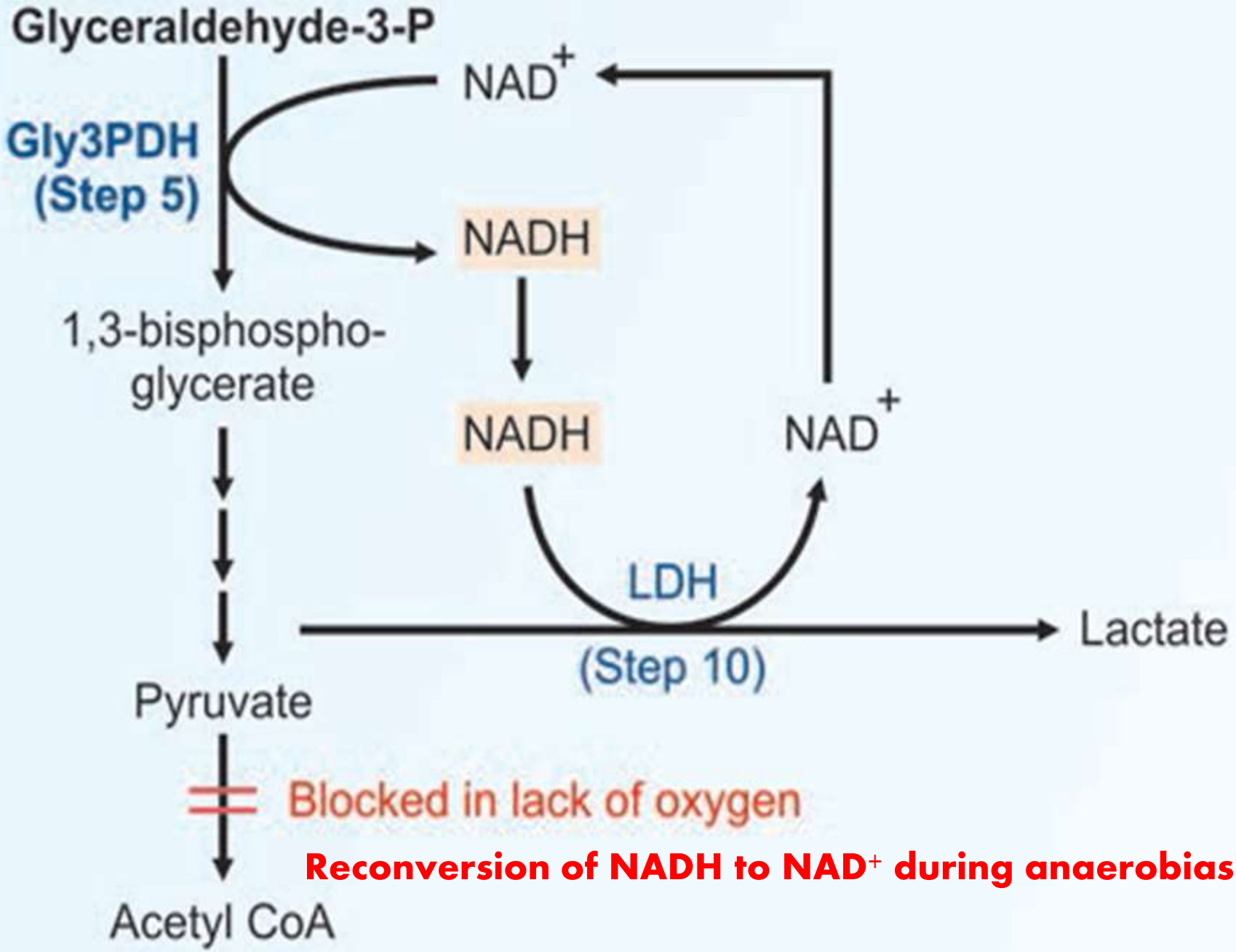
- In anaerobic condition, pyruvate is reduced to lactate by lactate dehydrogenase (LDH).
- LDH has 5 iso-enzymes.
- The cardiac iso-enzyme of LDH will be increased in myocardial infarcts.



Conversion of pyruvate to lactate

Significance of Lactate Production

- The NADH is obtained from the reaction catalysed by glyceraldehyde 3-phosphate dehydrogenase.
- The formation of lactate allows the regeneration of NAD^+ which can be reused by glyceraldehyde 3-phosphate dehydrogenase.
- Glycolysis proceeds even in the absence of oxygen to supply ATP.



Reconversion of NADH to NAD^+ during anaerobiasis

- **Glycolysis is very essential in skeletal muscle during strenous exercise where oxygen supply is very limited.**
- **In RBCs, there are no mitochondria.**
- **Glycolysis in the erythrocytes leads to lactate production**
- **RBCs derive energy only through glycolysis, where the end product is lactic acid.**

Lactic acidosis

- Elevation of lactic acid in the circulation (normal plasma 4-15 mg/dl) may occur due to its increased production or decreased utilization.
- Mild forms of lactic acidosis are associated with strenuous exercise, shock, respiratory diseases, cancers, low PDH activity, von Gierke's disease etc.

- **Severe forms of lactic acidosis are observed due to impairment/collapse of circulatory system -in myocardial infarction, pulmonary embolism, uncontrolled hemorrhage & severe shock.**
- **This type of lactic acidosis is due to inadequate supply of O_2 to the tissues with a drastic reduction in ATP synthesis, which may lead to death.**

- **Oxygen debt refers to the excess amount of O_2 required to recover.**
- **Measurement of plasma lactic acid is useful to know about the oxygen debt, and monitor the patient's recovery.**

Pasteur effect

- The inhibition of glycolysis by oxygen (aerobic condition) is known as Pasteur effect.
- Pasteur effect is due to the inhibition of the enzyme phosphofructokinase.
- Glycolytic intermediates from fructose 1,6-bisphosphate onwards decrease while the earlier intermediates accumulate.

Crabtree effect

- Inhibition of oxygen consumption by the addition of glucose to tissues having high aerobic glycolysis is known as Crabtree effect.
- Opposite to that of Pasteur effect.
- Crabtree effect is due to increased competition of glycolysis for inorganic phosphate (Pi) & NAD^+ which limits their availability for phosphorylation & oxidation.

References

- Textbook of Biochemistry – Harper
- Textbook of Biochemistry – Lippincott
- Textbook of Biochemistry – MN Chatterjea

Thank you